Sem 2 AY2425 MH1101 Calculus II Test (1 hour)

Name: .... **Matriculation Number:** ....

Tutorial Group: ....

Question 1 [14 marks]

Suppose that $a, b, g : \mathbb{R} \to \mathbb{R}$ are differentiable functions and f is integrable on any closed interval $[x_1, x_2]$ in $\mathbb{R}$.

(a) Find an expression for

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(t)\, dt$$

for $x \in \mathbb{R}$.

(b) Find an expression for

$$\frac{d}{dx} \int_{a(x)}^{b(x)} g(x) f(t)\, dt$$

(c) Now assume that $b : [0, \infty) \to \mathbb{R}$ is differentiable on $(0, \infty)$. If $b(x) \geq 0$ for all $x \geq 0$ and $b(0) = 1$, find $b(x)$ when

$$\frac{d}{dx} \int_{0}^{b(x)} t\, dt = 2x,$$

for all $x > 0$.

Answer.

More space for answers.

More space for answers.

Question 2 [18 marks]

(a) Find the area enclosed between the curves $y = \sqrt{x}$ and $y = x^4$ for $x \geq 0$.

(b) Use the disk-washer method to evaluate the volume of the solid generated by revolving the area enclosed between $y = \sqrt{x}$ and $y = x^4$ for $x \geq 0$ about the x-axis.

(c) Evaluate the volume in (b) using the method of cylindrical shells.

(d) We now consider the volume of the solid obtained by revolving the region enclosed between the x-axis and the curve $y = x - x^7$ from $x = 0$ to $x = 1$ about the y-axis.

Use either the disk-washer method *or* the method of cylindrical shells to calculate this volume and explain why the alternative method would not work.

Answer.

More space for answers.

More space for answers.

More space for answers.

Question 3 [18 marks]

(a) Find
$$\int \frac{1}{\sqrt{x} + x\sqrt{x}}\, dx$$

(b) Calculate the following definite integral when n is a positive integer:
$$\int_1^e x^n \ln x \, dx$$

(c) Evaluate the limit
$$\lim_{n\to\infty} \sum_{i=1}^{n} \frac{3i}{n^2} \sqrt{\frac{i}{n}}$$

Answer.

More space for answers.

More space for answers.

More space for answers.

Formulae Table

$\int \frac{1}{x}\,dx = \ln\lvert x\rvert + C$	$\int e^x\,dx = e^x + C$
$\int k\,dx = kx + C$	$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C\ (n \neq -1)$
$\int \sin x\,dx = -\cos x + C$	$\int \cos x\,dx = \sin x + C$
$\int \sec^2 x\,dx = \tan x + C$	$\int \csc^2 x\,dx = -\cot x + C$
$\int \sec x \tan x\,dx = \sec x + C$	$\int \csc x \cot x\,dx = -\csc x + C$
$\int \tan x\,dx = \ln\lvert\sec x\rvert + C$	$\int \sec x\,dx = \ln\lvert\sec x + \tan x\rvert + C$
$\int \frac{1}{1+x^2}\,dx = \tan^{-1} x$	

$\sin^2 x + \cos^2 x = 1$	$\tan^2 x + 1 = \sec^2 x$
$1 + \cot^2 x = \csc^2 x$	$\sin 2x = 2\sin x \cos x$
$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$	$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$

$$\sin A \cos B = \frac{1}{2}\left(\sin(A-B) + \sin(A+B)\right)$$
$$\sin A \sin B = \frac{1}{2}\left(\cos(A-B) - \cos(A+B)\right)$$
$$\cos A \cos B = \frac{1}{2}\left(\cos(A-B) + \cos(A+B)\right)$$

$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$
$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$
$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$	$\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$

– End of Test –